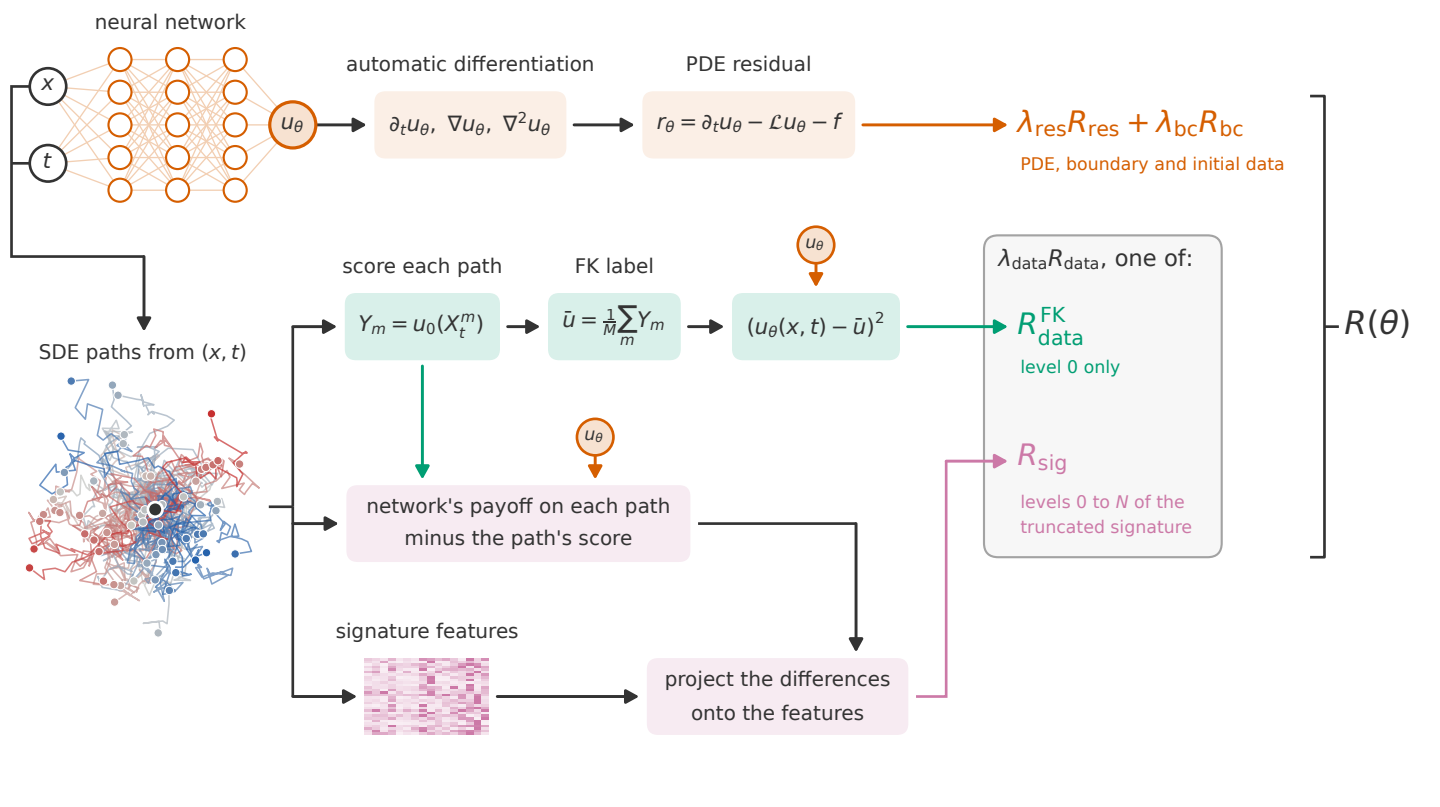

neural network
x
t
u_θ
automatic differentiation
$\partial_t u_\theta,\ \nabla u_\theta,\ \nabla^2 u_\theta$
PDE residual
$r_\theta = \partial_t u_\theta - \mathcal{L} u_\theta - f$
$\lambda_{\text{res}} R_{\text{res}} + \lambda_{\text{bc}} R_{\text{bc}}$
PDE, boundary and initial data
$R(\theta)$
SDE paths from (x, t)
score each path
$Y_m = u_0(X_t^m)$
FK label
$\bar{u} = \frac{1}{M}\sum_m Y_m$
u_θ
$(u_\theta(x,t) - \bar{u})^2$
$\lambda_{\text{data}} R_{\text{data}}$, one of:
$R_{\text{data}}^{\text{FK}}$
level 0 only
R_{sig}
levels 0 to N of the truncated signature
u_θ
network's payoff on each path minus the path's score
signature features
project the differences onto the features